

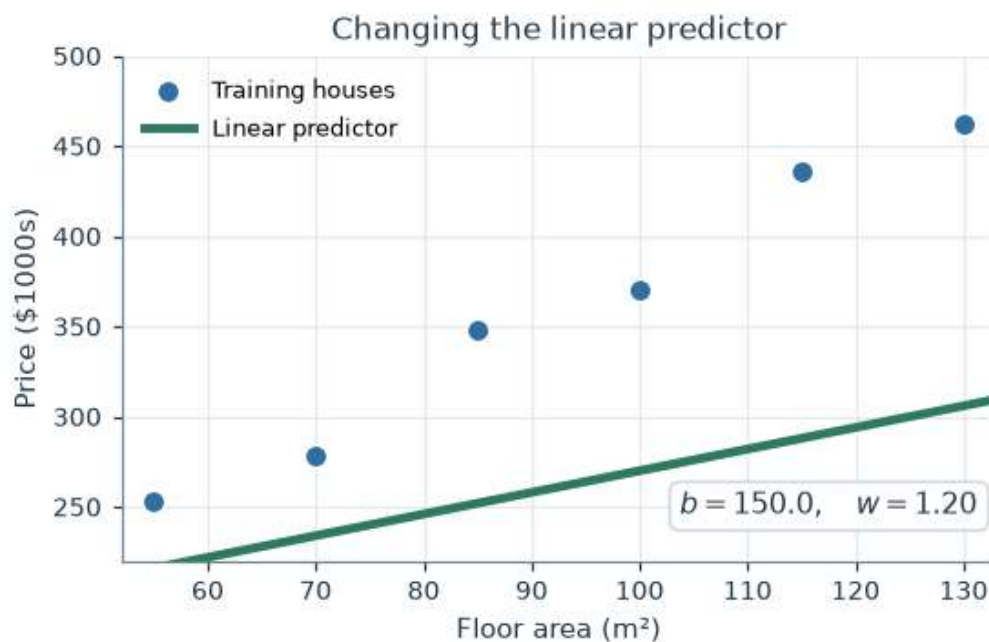
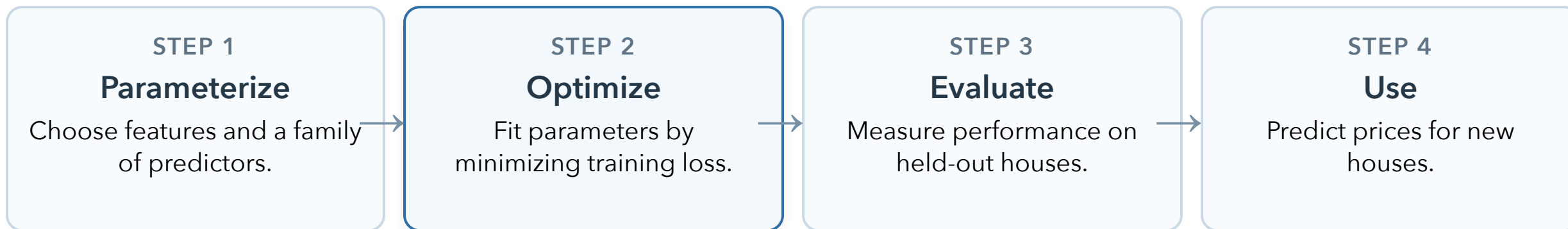
CMPUT 267: Machine Learning I · Fall 2026

Optimization Basics and Closed Form Linear Regression

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Recall: A Practical ML Pipeline



Recall: ERM Has Two Steps

For a fixed dataset $\mathcal{D}$, empirical risk minimization chooses

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_{\mathcal{D}}(f), \quad \hat{L}_{\mathcal{D}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i), y_i).$$

Estimation

The data construct the objective $\hat{L}_{\mathcal{D}}$.

Optimization

Search $\mathcal{F}$ for a predictor that minimizes the objective.

Today's Focus: Optimization

Optimization: Find the best choice from a set of allowed choices, according to a numerical objective

For ERM:

Choices

$$f \in \mathcal{F}$$

Candidate predictors

Objective

$$\hat{L}_{\mathcal{D}}(f)$$

Empirical risk

Best Choice

$$\hat{f}$$

A predictor with minimum empirical risk

Minimum Value and Minimizer

Let $g : \mathcal{W} \rightarrow \mathbb{R}$ be an objective function.

Minimum Value

$$\min_{w \in \mathcal{W}} g(w)$$

The smallest output of g .

Minimizer

$$w^* \in \arg \min_{w \in \mathcal{W}} g(w)$$

An input where the minimum occurs.

$$\min_{w \in \mathcal{W}} g(w) = g(w^*).$$

Example: Concrete Minimum

$$g(w) = (w - 2)^2 + 1$$

Minimizer

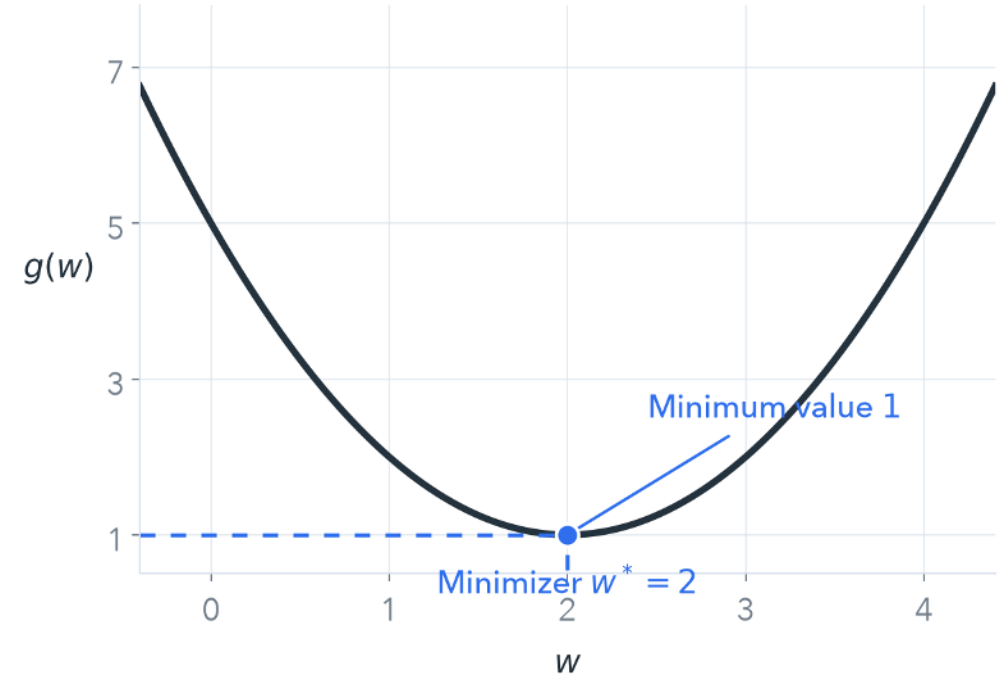
$$w^* = 2$$

The input coordinate of the bottom of the parabola.

Minimum Value

$$g(w^*) = 1$$

The objective value at the bottom.



The Domain Changes the Answer

Use the same objective $g(w) = w^2$ with two different domains.

Continuous Domain

$$\mathcal{W} = \mathbb{R}$$

$$w^* = 0 \text{ and } g(w^*) = 0$$

Two Allowed Values

$$\mathcal{W} = \{-1, 2\}$$

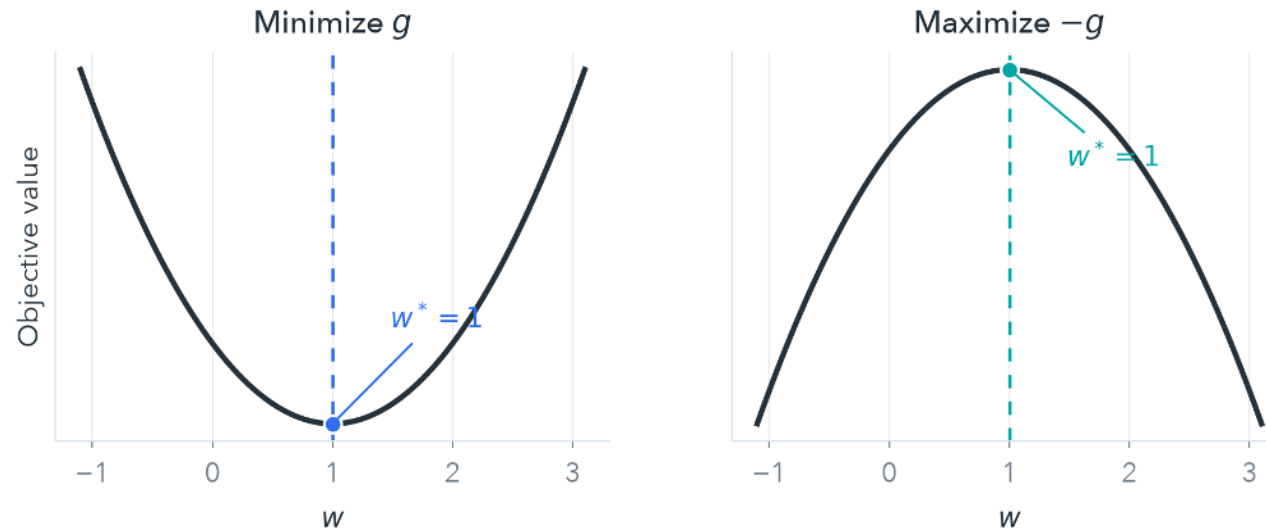
$$w^* = -1 \text{ and } g(w^*) = 1$$

 An optimization problem includes both the objective g and the domain $\mathcal{W}$.

Minimization and Maximization

A Maximization problem can be cast into a minimization problem.

Negating an minimization problem:



$$\arg \min_{w \in \mathcal{W}} g(w) = \arg \max_{w \in \mathcal{W}} (-g(w)), \quad \min_{w \in \mathcal{W}} g(w) = - \max_{w \in \mathcal{W}} (-g(w)).$$

Optimizer and Optimal Value

Let $g(w) = (w - 1)^2 + 1$. Negating g leaves the optimizer at the same horizontal location.

Minimize g

$$\arg \min g(w) = 1$$

The minimum value is $g(1) = 1$.

Maximize $-g$

$$\arg \max [-g(w)] = 1$$

The maximum value is $-g(1) = -1$.

Optimization: Discrete Domains

Let $g(w) = w^2$ and $\mathcal{W} = \{-2, -1, 2, 3\}$.

Allowed input w	-2	-1	2	3
Objective value $g(w)$	4	1	4	9

$$w^* = -1, \quad \min_{w \in \mathcal{W}} g(w) = 1.$$



With no additional structure, a discrete search may need to evaluate every allowed input.

Discrete and Continuous Domains

The domain determines which tools are available.

Discrete Domain

Evaluate allowed candidates and compare their objective values.

Continuous Domain

When the objective is smooth enough, use derivatives as a tool.

 This course focuses on continuous parameter spaces such as $\mathbb{R}^d$.

Goal: Optimize Linear Regression with Squared Loss

Recall the house-price problem with size s and number of rooms r :

$$f(s, r) = b + w_s s + w_r r,$$
$$\ell(f(s_i, r_i), y_i) = (f(s_i, r_i) - y_i)^2 = (b + w_s s_i + w_r r_i - y_i)^2.$$

Across all n training houses, learning means minimizing the empirical risk:

$$\hat{L}_{\mathcal{D}}(b, w_s, w_r) = \frac{1}{n} \sum_{i=1}^n (b + w_s s_i + w_r r_i - y_i)^2,$$
$$(\hat{b}, \hat{w}_s, \hat{w}_r) \in \arg \min_{(b, w_s, w_r) \in \mathbb{R}^3} \hat{L}_{\mathcal{D}}(b, w_s, w_r).$$

 This objective is convex in (b, w_s, w_r) . We will use that property before deriving its solution.

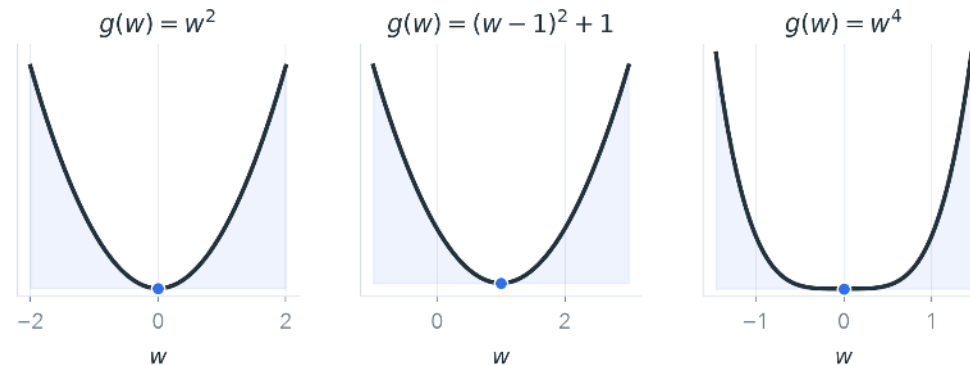
Convex

Assume $g : \mathcal{W} \rightarrow \mathbb{R}$ is twice differentiable.

Convex

A function g is convex on $\mathcal{W}$ if
 $g''(w) \geq 0$ for all $w \in \mathcal{W}$.

Intuitively, a convex function is **bowl shaped**.



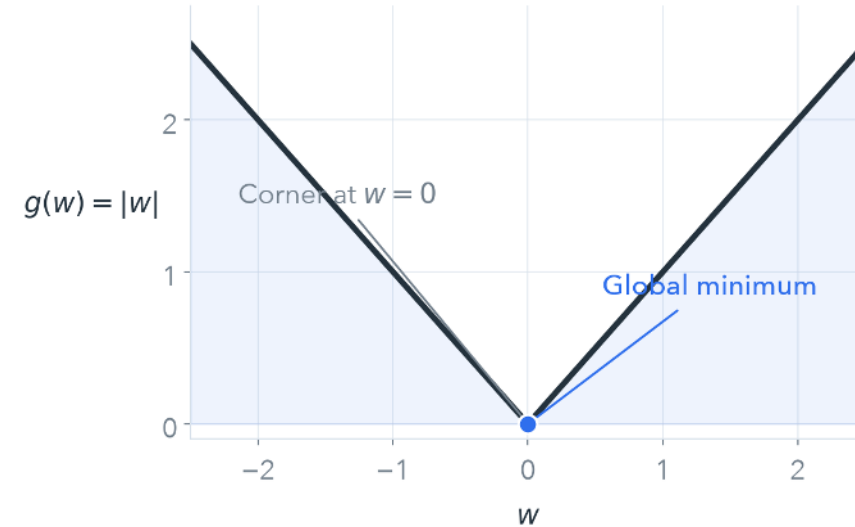
Caveat: A Nonsmooth Convex Function

A convex function does not always mean a smooth function.

The absolute-value

$$g(w) = |w|$$

is convex and has a global minimum at $w = 0$.

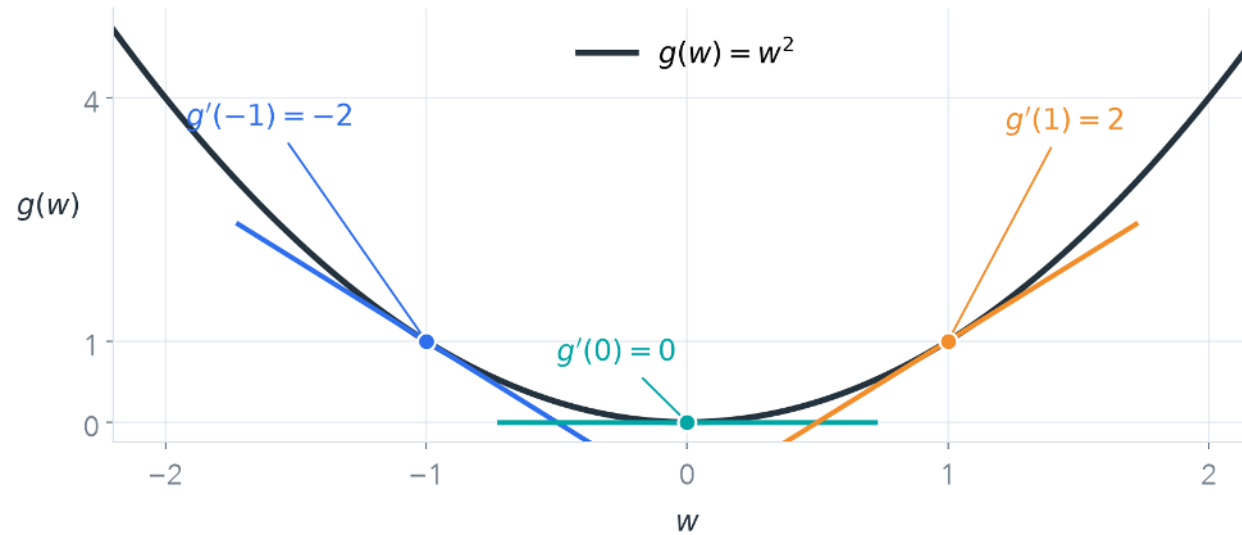


However, $g'(0)$ does not exist.

For this lecture, we stay with twice-differentiable objectives.

Derivatives Describe the Shape

For $g(w) = w^2$, the derivative is the slope of the curve at w .



$$g'(-1) = -2, \quad g'(0) = 0, \quad g'(1) = 2.$$

i The slope changes from negative to zero to positive as we pass through the minimum.

Minimizing a Convex Function

Assume g is convex and differentiable. Convexity makes finding a minimizer easier because every stationary point is globally optimal.

1 Domain $\mathcal{W} = \mathbb{R}$

Minimizer w^* satisfies $g'(w^*) = 0$.

If no solution exists, g may have no minimizer on $\mathbb{R}$.

2 Domain $\mathcal{W} = [a, b]$

Check for $g'(w^*) = 0$ with $w^* \in [a, b]$.

If there does not exist w^* in $[a, b]$ satisfying $g'(w^*) = 0$, compare $g(a)$ and $g(b)$.

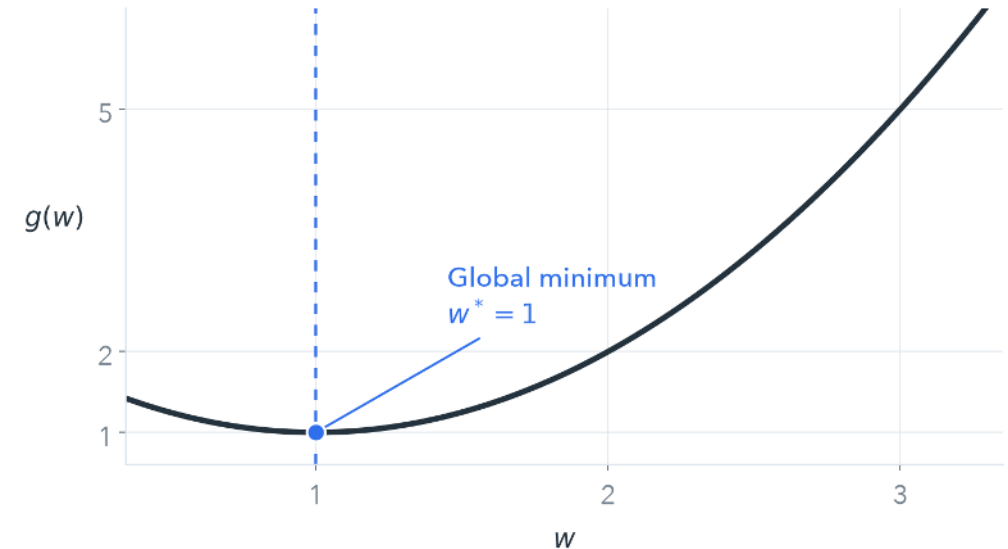
A Convex Objective on $\mathbb{R}$

For $g(w) = w^2 - 2w + 2$,

$$g'(w) = 2w - 2, \quad g''(w) = 2 \geq 0.$$

The nonnegative second derivative confirms convexity. The minimizer solves

$$\begin{aligned} g'(w) = 0 &\implies 2w - 2 = 0 \\ &\implies w^* = 1. \end{aligned}$$



A Convex Objective on an Interval

Now minimize $g(w) = w^2 - 2w + 2$ over $\mathcal{W} = [2, 3]$.



The stationary point $w = 1$ lies outside the domain, so compare the endpoints:

$$g(2) = 2, \quad g(3) = 5, \quad w^* = 2.$$

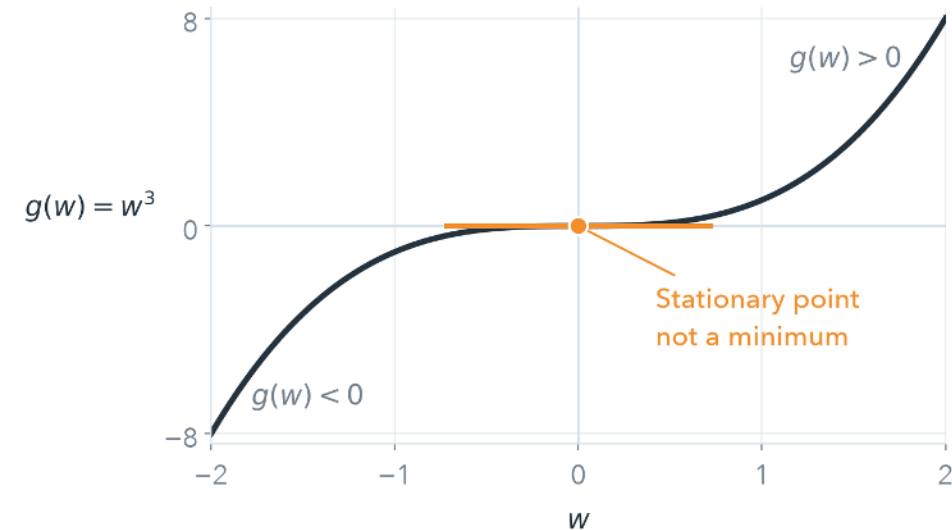
Why Convexity Matters

For $g(w) = w^3$,

$$g'(w) = 3w^2, \quad g''(w) = 6w.$$

Solving $g'(w) = 0$ gives $w = 0$.

But $g''(w) < 0$ for $w < 0$, so g is not convex.



 A zero derivative identifies a stationary point. Without convexity, it need not identify a global minimum.

Optimization in Several Dimensions

Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ take a parameter vector

$$\mathbf{w} = (w_1, \dots, w_d)^\top.$$

For a convex, differentiable objective, a minimizer satisfies

$$\frac{\partial g}{\partial w_j}(\mathbf{w}^*) = 0 \quad \text{for every } j \in \{1, \dots, d\}.$$

 These d equations must be solved together because each partial derivative may depend on several coordinates of $\mathbf{w}$.

Convexity in Several Dimensions

The gradient collects all first partial derivatives:

$$\nabla g(\mathbf{w}) = \begin{bmatrix} \partial g / \partial w_1 \\ \vdots \\ \partial g / \partial w_d \end{bmatrix}.$$

The Hessian $\nabla^2 g(\mathbf{w})$ collects the second partial derivatives.

i If the Hessian is positive semidefinite everywhere, then g is convex. In this course, we will usually establish convexity for you.

A Two-Dimensional Example

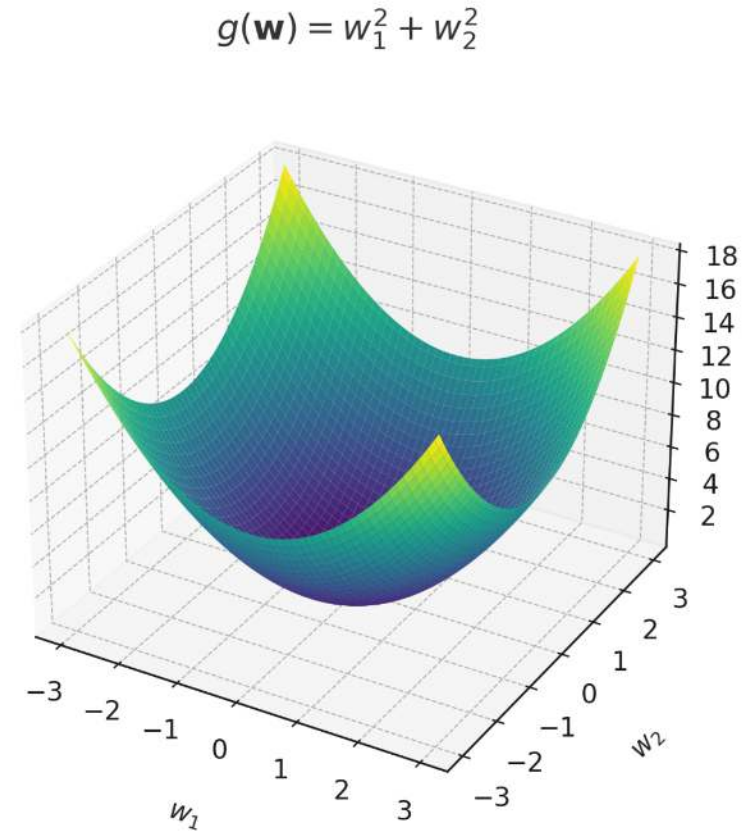
Let

$$g(\mathbf{w}) = w_1^2 + w_2^2.$$

Then

$$\frac{\partial g}{\partial w_1} = 2w_1, \quad \frac{\partial g}{\partial w_2} = 2w_2.$$

$$\mathbf{w}^* = (0, 0)^\top.$$



Coupled Partial Derivatives

Consider the convex objective

$$g(w_1, w_2) = (w_1 + w_2 - 3)^2 + (w_1 - 1)^2.$$

$$\frac{\partial g}{\partial w_1} = 4w_1 + 2w_2 - 8, \quad \frac{\partial g}{\partial w_2} = 2w_1 + 2w_2 - 6.$$

$$2w_1 + w_2 = 4, \quad w_1 + w_2 = 3 \quad \implies \quad \mathbf{w}^* = (1, 2)^\top.$$



Each equation contains both parameters, so the equations must be solved as a system.

Closed Form Solution for Linear Regression

Linear Regression as a Function Class

ERM optimizes over a set of functions:

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{L}_{\mathcal{D}}(f).$$

In linear regression, every predictor in $\mathcal{F}$ has the same form:

$$f_{\mathbf{w}}(\mathbf{x}) = b + w_1 \underbrace{s}_{\text{size}} + w_2 \underbrace{r}_{\text{number of rooms}}.$$

The predictors differ through

$$\mathbf{w} = (b, w_1, w_2)^{\top}.$$

 We call $\mathbf{w}$ the **parameters** or **weights**. Choosing a predictor from $\mathcal{F}$ means choosing these values.

The Bias Term

In

$$f_{\mathbf{w}}(\mathbf{x}) = b + w_1 s + w_2 r,$$

the constant b is called the **bias**. It shifts every prediction by the same amount.

$$\underbrace{w_1 s + w_2 r}_{\text{must predict 0 when } s=r=0} \longrightarrow \underbrace{b + w_1 s + w_2 r}_{\text{can shift up or down}}$$

 The bias lets the data determine the baseline prediction.

The Bias as an Extra Weight

Set $x_0 = 1$ and rename the bias w_0 :

$$\tilde{\mathbf{x}} = (1, s, r)^\top, \quad \tilde{\mathbf{w}} = (w_0, w_1, w_2)^\top.$$

$$f_{\tilde{\mathbf{w}}}(\tilde{\mathbf{x}}) = 1 \cdot w_0 + sw_1 + rw_2 = \tilde{\mathbf{x}}^\top \tilde{\mathbf{w}}.$$

i From here onward, we drop the tildes and assume that the first feature is $x_0 = 1$.

ERM over the Weight Vector

For the linear function class $\mathcal{F} = \{f_{\mathbf{w}} : f_{\mathbf{w}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}, \mathbf{w} \in \mathbb{R}^{d+1}\}$,

Recall ERM objective:

$$\hat{\mathbf{w}} \in \arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \hat{L}_{\mathcal{D}}(\mathbf{w}), \quad \hat{f}(\mathbf{x}) = \mathbf{x}^\top \hat{\mathbf{w}}.$$

Here we use squared loss: $\ell(\hat{y}, y) = (\hat{y} - y)^2$.

The empirical risk becomes

$$\hat{L}_{\mathcal{D}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^\top \mathbf{w} - y_i)^2.$$

 As a function of $\mathbf{w}$, this objective is convex and twice differentiable.

Goal: Solve for the Minimizer

We want to solve

$$\hat{\mathbf{w}} \in \arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \hat{L}_{\mathcal{D}}(\mathbf{w}), \quad \hat{L}_{\mathcal{D}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^{\top} \mathbf{w} - y_i)^2.$$

The objective is convex and differentiable. From the earlier convexity result, its minimizer satisfies

$$\boxed{\nabla \hat{L}_{\mathcal{D}}(\hat{\mathbf{w}}) = \mathbf{0}}.$$

i The rest of the derivation solves this equation for $\hat{\mathbf{w}}$.

Step 1: Expand the Objective

Write the squared-loss objective in terms of the coordinates of $\mathbf{w}$:

$$\begin{aligned}\hat{L}_{\mathcal{D}}(\mathbf{w}) &= \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{x}_i^{\top} \mathbf{w}, y_i) \\ &= \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^{\top} \mathbf{w} - y_i)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_{i0}w_0 + \cdots + x_{id}w_d - y_i)^2.\end{aligned}$$

Name the loss from one training example:

$$\hat{L}_i(\mathbf{w}) = (x_{i0}w_0 + \cdots + x_{id}w_d - y_i)^2, \quad \hat{L}_{\mathcal{D}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \hat{L}_i(\mathbf{w}).$$

Step 2: One Coordinate

Recall the decomposition from the previous slide:

$$\hat{L}_{\mathcal{D}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \hat{L}_i(\mathbf{w}), \quad \hat{L}_i(\mathbf{w}) = (x_{i0}w_0 + \cdots + x_{id}w_d - y_i)^2.$$

For one example i , differentiate $\hat{L}_i$ with respect to one coordinate w_j :

$$\begin{aligned} \frac{\partial \hat{L}_i}{\partial w_j}(\mathbf{w}) &= 2(x_{i0}w_0 + \cdots + x_{id}w_d - y_i) \underbrace{\frac{\partial}{\partial w_j}(\cdots + x_{ij}w_j + \cdots)}_{x_{ij}} \\ &= 2x_{ij}(x_{i0}w_0 + \cdots + x_{id}w_d - y_i). \end{aligned}$$

Step 3: Derivative of the Average

Now return to $\hat{L}_{\mathcal{D}} = \frac{1}{n} \sum_i \hat{L}_i$. Differentiation is linear:

$$\begin{aligned} \frac{\partial \hat{L}_{\mathcal{D}}}{\partial w_j}(\mathbf{w}) &= \frac{1}{n} \sum_{i=1}^n \frac{\partial \hat{L}_i}{\partial w_j}(\mathbf{w}) \\ &= \frac{2}{n} \sum_{i=1}^n x_{ij} (x_{i0}w_0 + \cdots + x_{id}w_d - y_i). \end{aligned}$$

i This gives the derivative with respect to one coordinate w_j . We repeat it for $j = 0, \dots, d$.

Step 4: Zero Derivatives

At the minimizer, the derivative equals zero for every coordinate w_j :

$$0 = \frac{\partial \hat{L}_{\mathcal{D}}}{\partial w_j}(\hat{\mathbf{w}}) = \frac{2}{n} \sum_{i=1}^n x_{ij} (x_{i0}\hat{w}_0 + \cdots + x_{id}\hat{w}_d - y_i).$$

This equation holds for every $j = 0, \dots, d$.

Cancel the nonzero constant $2/n$:

$$\sum_{i=1}^n x_{ij} (x_{i0}\hat{w}_0 + \cdots + x_{id}\hat{w}_d - y_i) = 0.$$

Step 5: Rearrangement

For a fixed coordinate j , distribute x_{ij} and separate the terms:

$$\left(\sum_{i=1}^n x_{ij}x_{i0}\right)\hat{w}_0 + \cdots + \left(\sum_{i=1}^n x_{ij}x_{id}\right)\hat{w}_d - \sum_{i=1}^n x_{ij}y_i = 0.$$

Move the label term to the other side:

$$\left(\sum_{i=1}^n x_{ij}x_{i0}\right)\hat{w}_0 + \cdots + \left(\sum_{i=1}^n x_{ij}x_{id}\right)\hat{w}_d = \sum_{i=1}^n x_{ij}y_i.$$

This is one linear equation in $\hat{w}_0, \dots, \hat{w}_d$. There is one such equation for each j .

Step 6: Matrix

Stack the coordinate equations.

Let X have rows $\mathbf{x}_i^\top$, and let $\sum_i = \sum_{i=1}^n$:

$$\underbrace{\begin{bmatrix} \sum_i x_{i0}x_{i0} & \cdots & \sum_i x_{i0}x_{id} \\ \vdots & \ddots & \vdots \\ \sum_i x_{ij}x_{i0} & \cdots & \sum_i x_{ij}x_{id} \\ \vdots & \ddots & \vdots \\ \sum_i x_{id}x_{i0} & \cdots & \sum_i x_{id}x_{id} \end{bmatrix}}_{X^\top X} \begin{bmatrix} \hat{w}_0 \\ \vdots \\ \hat{w}_j \\ \vdots \\ \hat{w}_d \end{bmatrix} = \underbrace{\begin{bmatrix} \sum_i x_{i0}y_i \\ \vdots \\ \sum_i x_{ij}y_i \\ \vdots \\ \sum_i x_{id}y_i \end{bmatrix}}_{X^\top \mathbf{y}}.$$

Each (j, k) entry of $X^\top X$ is $\sum_i x_{ij}x_{ik}$. The blue row is the equation for coordinate w_j .

Step 7: Normal Equations

The labels under the expanded matrix give its compact form:

$$X^{\top} X \hat{\mathbf{w}} = X^{\top} \mathbf{y}.$$

Define $A = X^{\top} X$ and $\mathbf{b} = X^{\top} \mathbf{y}$. Then the same system is

$$A \hat{\mathbf{w}} = \mathbf{b}.$$

Step 8: Closed Form

If $A = X^\top X$ is invertible, multiply the normal equations by A^{-1} :

$$\hat{\mathbf{w}} = A^{-1} \mathbf{b} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

- i* Convexity guarantees that this solution is a global minimizer of the empirical risk.
- i* The formula is closed form because a finite sequence of specified operations produces the exact minimizer.

Result: The Closed Form Learner

Given

$$\mathcal{D} = ((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)),$$

the learner performs the following calculations:

$$\begin{aligned} X &\leftarrow \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}, & \mathbf{y} &\leftarrow \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \\ A &\leftarrow X^\top X, & \mathbf{b} &\leftarrow X^\top \mathbf{y}, \\ \hat{\mathbf{w}} &\leftarrow A^{-1} \mathbf{b}, \\ \hat{f}(\mathbf{x}) &\leftarrow \mathbf{x}^\top \hat{\mathbf{w}}. \end{aligned}$$

A Two-Feature House Dataset

Use size s and number of rooms r to predict price in \$1000s. Prepend 1 for the bias:

Size s_i	Rooms r_i	Price y_i	Augmented feature $\mathbf{x}_i$
100	2	300	$(1, 100, 2)^\top$
120	4	400	$(1, 120, 4)^\top$
150	3	420	$(1, 150, 3)^\top$
180	5	550	$(1, 180, 5)^\top$

$$X = \begin{bmatrix} 1 & 100 & 2 \\ 1 & 120 & 4 \\ 1 & 150 & 3 \\ 1 & 180 & 5 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 300 \\ 400 \\ 420 \\ 550 \end{bmatrix}.$$

Compute A and $\mathbf{b}$

For these four training examples,

$$A = \begin{bmatrix} 4 & 550 & 14 \\ 550 & 79,300 & 2,030 \\ 14 & 2,030 & 54 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1,670 \\ 240,000 \\ 6,210 \end{bmatrix}.$$

These are exactly

$$A = X^\top X, \quad \mathbf{b} = X^\top \mathbf{y}.$$

The Fitted House-Price Predictor

$$\hat{\mathbf{w}} = A^{-1}\mathbf{b} \approx \underbrace{\begin{bmatrix} 44.01 \\ 1.84 \\ 34.29 \end{bmatrix}}_{(\hat{b}, \hat{w}_s, \hat{w}_r)^\top}, \quad \hat{f}(s, r) = 44.01 + 1.84s + 34.29r.$$

Size s_i	Rooms r_i	True price y_i	Prediction $\hat{f}(s_i, r_i)$
100	2	300	296.94
120	4	400	402.38
150	3	420	423.40
180	5	550	547.28

i The fitted empirical risk is approximately 8.50.

When the Formula Needs Care

A Is Invertible

The formula $A^{-1}\mathbf{b}$ gives a unique minimizer.

A Is Singular

The inverse does not exist. A pseudoinverse can select a least-squares solution.

i In numerical code, solve the system $A\hat{\mathbf{w}} = \mathbf{b}$ directly instead of explicitly forming A^{-1} .

Closed form is exact, but matrix operations can become expensive when the number of features is large.

Wrap-up

- An optimization problem specifies an objective and a domain.
- $\min$ gives the best objective value. $\arg \min$ gives the inputs that achieve it.
- Convexity makes stationary points globally meaningful.
- Parameterization turns optimization over predictors into optimization over weight vectors.
- Squared-loss linear regression produces the normal equations $A\hat{\mathbf{w}} = \mathbf{b}$.
- When A is invertible, $\hat{\mathbf{w}} = A^{-1}\mathbf{b}$ is the closed form solution.

 Next: why closed form solutions are not always available, and how gradient descent approximates a minimizer.